Physics and Astronomy (joint degree)

Vrije Universiteit Amsterdam - Faculty of Science - Physics and Astronomy (joint degree) - 2025-2026

Physics and Astronomy is a joint degree master's programme of University of Amsterdam and VU Amsterdam. For further information about the programme go to the UvA Course Catalogue:

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-opleiding/opleiding/8800

Info

Level	Master
Language	English
Duration	2 years
Type	Full Time
Credits	120 EC
Faculty	Faculty of Science

Joint degree
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Physics and Astronomy JD Compulsory courses taught at VU

Courses

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Light-Tissue Interaction	P1	6EC	X_428572
Current Sustainable Energy Technologies	P3	6EC	X_422582

Physics and Astronomy JD Optional Courses taught at VU

Courses

Course Name	Period	Credits	Code
BioSolar Cells	P1	6EC	X_428531
From Physics to Physiology	P1	6EC	X_430112
Managing Science and Technology in Society	P1	6EC	AM_470586
Research Methods for Analyzing Complex Problems	P1	6EC	AM_1182
Science and Communication	P1	6EC	AM_470587
Advanced Medical Image Processing	P2	6EC	X_422610
Communication, Organization and Management	P2	6EC	AM_470572
Management of Sustainable Innovation	P2	6EC	E_DBI_MANSI
Science Journalism	P2	6EC	AM_471014
Soft Condensed Matter and Biological Physics	P2	6EC	X_420167
Computational Optical Imaging	P3	3EC	XMU_0052
Laser Cooling and Trapping of Atoms and Molecules	P3	3EC	XMU_0070
Science in Dialogue	P3	6EC	AM_1002
Tutoring Students	P3	3EC	X_432625
Scientific Writing in English	P3, P6	3EC	X_400592
Deep Learning for Medical Image Analysis	P4	6EC	XM_0151
Entrepreneurship in Science and Technology	P4	6EC	XMU_0066
Materials for Light-Energy Conversion	P4	3EC	XMU_0071
Parameter Estimation Applied to Medical and Biological Sciences	P4	6EC	X_432631
Ultrafast Laser Physics	P4	6EC	XMU_0035
Science and Society in Historical Perspective	P4+5	6EC	X_400424
Advanced 3D and 4D Medical Imaging	P5	6EC	X_428569
Advanced Biophysics	P5	6EC	X_422583
Photosynthesis and Energy	P5	6EC	X_422553
Tracer Kinetic Modelling	P5	6EC	XM_0014
Electronics and Signal Processing	P6	6EC	XMU_0069

Physics and Astronomy expired courses

Courses

Advanced 3D and 4D Medical Imaging

Course Code	X_428569
Credits	6

Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I.H.M. van Stokkum
Examiner	
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130455

Additional Information Teaching Methods

Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Advanced Biophysics

Course Code	X_422583
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D. Dulin
Examiner	dr. D. Dulin
Teaching Staff	dr. D. Dulin, dr. ir. I. Heller
Teaching method(s)	Lecture

Course Objective

Learning objectives

- Understand the physics underlying the biological problems and biophysical research methods featured in this course
- Be able to explain how these methods can be exploited to address specific biological problems and identify strengths and weaknesses of methods in biological context
- Critically reflect on your understanding of methods application in assigned literature and formulate relevant questions
- Analyze scientific literature that uses the discussed methodologies to addresses biological problems and reflect on suitability of used methods
- Communicate and present your literature analysis and critically reflect on analyses by other students

Course Content

Life is, by its very definition, a dynamic quantity. In this course an overview is given of current biophysics, biophotonics, and biomedical research fields that investigate the dynamic processes that take place in biomolecules, membranes, cells and tissues and are indispensable for biological functionality. This course aims to highlight the 'what', 'why' and 'how' of these research fields: what are the established and new biophysical insights, why are these insights important and interesting, and how are these results being obtained.

The following biophysical/biomedical topics and techniques are featured in this course:

- DNA processing and dynamics
  o Topics: DNA structure and stability, DNA/RNA polymerase, DNA architectural proteins, DNA repair.
  o Techniques: optical tweezers, AFM, tethered particle motion, magnetic tweezers, superresolution imaging.
- Protein dynamics
  o Topics: Photoactive proteins, light-driven enzymes, Motor proteins, optogenetics.
  o Techniques: infrared spectroscopy, time-resolved spectroscopy, laser techniques
- Membrane dynamics
  o Topics: Photosynthesis, membrane protein diffusion, Neuroimaging.
  o Techniques: AFM, single molecule, nonlinear imaging, polarized spectroscopy.
- Cell/tissue dynamics
  o Topics: Optical and/or mechanical characterization of tissue or cells.
  o Techniques: (Fluorescence) microscopy, optical coherence microscopy, mechanical analysis.

Additional Information Teaching Methods

Lectures, guest lectures, literature essay and assignments, oral literature presentation

Method of Assessment

- Literature assignments
- Essay (literature or research proposal) (70% final grade)
- Essays from two other students peer review (10% final grade)
- Oral literature presentation (20% final grade)

Entry Requirements

BSc. Physics, BSc. Medical Natural Sciences, BSc Chemistry or comparable.

For MNS students to participate in this course, it is advised to have successfully completed the course Microscopie en spectroscopie voor MNW (X_420586) and Mechanics and Thermodynamics in the Cell (X_422589).

Literature

Notes, handouts and papers.

Additional Information Target Audience

mMNS-PoL, mPhys-LSBP, mPhys-PLH

Advanced Medical Image Processing

Course Code	X_422610
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. M.M. Yaqub
Examiner	dr. ir. M.M. Yaqub
Teaching Staff	dr. ir. A.M. Wink, dr. ir. M.M. Yaqub
Teaching method(s)	Computer lab, Written partial exam, Lecture

Course Objective

1. learning the difference between experimental environments such as Matlab and Python, and industry environments such as C++
2. learning the possibilities of C++ for creating classes and processing methods
3. applying these techniques in advanced medical imaging analysis using one or more of the following topics: data structures, interpolation, connected component algorithms, geometric modelling, optimized algorithms, noise filtering, wavelet decomposition, machine learning.
4. developing image analysis software tools from concept to algorithm in C++ .

Course Content

Software for computationally intensive tasks such as dense signal and image analysis, deep learning and big data applications is usually written in C++. Even for specialist data science, mathematical and statistical software such as Python, Matlab and R, the underlying libraries that ensure optimal efficiency, the low-level processing is usually implemented as C++ libraries.

This means that knowledge of C++ is often a requirement for working in these fields of science. The aim of this course is to learn the development of medical image processing methods in C++ by using modern language constructs such as classes, standard libraries and templates. It demonstrates the importance of program syntax and structure to achieve optimal speed, efficiency and maintainability.

Additional Information Teaching Methods

lectures, practice sessions, reviews

Method of Assessment

Programming assignment on a specific topic in medical image processing, including evaluation of the program and reporting on the methods and results. The assignment will be scored at 50% of the final score. Written exam which is accessible after completing the weekly sessions and the individual assignment. The written exam will be scored at 50% of the final score. Note that both assignment score and written exam score need to be more than 55% individually.

Additional Information Target Audience

MNS-master & Master Physcics of Life & Health

Recommended background knowledge

- Medical Imaging (Bachelor physics and bachelor medical natural sciences)
- Image processing for MNS (Inleiding medische beeldbewerking).
- Some experience with programming, e.g. MatLab.

BioSolar Cells

Course Code	X_428531
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.N. Frese
Examiner	dr. R.N. Frese
Teaching Staff	dr. R.N. Frese
Teaching method(s)	Lecture

Course Objective

The course provides a broad overview of BioSolar Cells scientific research. It serves to acquire detailed knowledge of parts that comprise biological, chemical and physical materials approaches to utilize sunlight for fuel and or capture carbon dioxide based on biological or artificial photosynthesis.

Focus is on technologies that are being researched right now and are not yet implemented at full scale. Part of the course is dedicated to technology transition theory, focusing on the hurdles and bottlenecks for lab-based researched to be implemented. Important skills that are developed are the assessment of technologies that are not yet implemented at scale for current decision makers in renewable energy.

In the course, students will be reading scientific papers from the exact sciences and other academics, information leaflets for different target audiences on renewable energy efforts. Subsequently, students will be following lectures on focused topics and learn to distill relevant information from them. In work, students aim to present an overview of specific technology using scientific and other references with an assessment of the relevance of references used.

Combined this course will allow students to investigate a topic of their own interest and to acquire the necessary

skills of assessing and presenting scientific information. The course provides the basic skills for a successful master study program.

The course consists of

- a group assignment on a relevant issue related to Biosolar Cells, presentation
- a literature study on one of the themes from BioSolar Cells, using a template geared to fast and critical information transfer
- a set of questions and annotations in a diverse set of reading materials

Course Content

The course consists of three parts, and starts with interactive lectures by experts on each of the three themes from BioSolar Cells. Students will read one or two papers before each lecture and formulate questions that have been researched by themselves first. In the second stage of this course, the students first work out a group assignment (groups of 3-4 students) leading to a team presentation on a specific topic, and then conduct an individual assessment on a BioSolar-Cells related topic of their own choice and present their work in the form of a scientific review paper where the specific topic is compared with other solutions, the paper concludes with a brief description of a research proposal.

Additional Information Teaching Methods

- Lecture
- Self-study

Method of Assessment

Grade is a weighted sum of the presentation and the report.

For the quality of the questions submitted prior to the lectures the student will receive the judgement “insufficient”, “sufficient” or “excellent” by the lecturer and a +/- bonus will be added/subtracted from the grade depending on the final sum.

Also for the quality of the referee reports the student will receive the judgement “insufficient”, “sufficient” or “excellent” and +/- bonus point. Maximum grade is always 10.

A retake is only allowed once and only when all materials have been submitted with requested content and within the prescribed deadline.

Entry Requirements

Physics or chemistry at bachelor level or a mixture of these at bachelor level supplemented with other topics. Highschool biology is recommended. For those without prior knowledge on photosynthesis an introductory lecture and reading material will be provided.

Literature

During the course a diverse set of scientific papers, leaflets, academic papers, brochures, and other materials is provided for reading and annotating (compulsory)

Additional Information Target Audience

master students physics, chemistry, SBI, lifesciences

Additional Information

This course can be challenging for those that do not read much.

Explanation Canvas

Information, materials, rubrics and templates are on canvas, provided on a weekly basis. Students are expected to read all materials without specific instructions from a teacher.

Recommended background knowledge

for those without prior knowledge on photosynthesis an introductory lecture will be given

Communication, Organization and Management

Course Code	AM_470572
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C.A.C.M. Pittens MSc
Examiner	dr. C.A.C.M. Pittens MSc
Teaching Staff	dr. S.S. Rai, dr. R.M.H. Peters, dr. C.A.C.M. Pittens MSc
Teaching method(s)	Study Group, Lecture

Course Objective

After completion of this course, you will be able to:

- identify and discuss knowledge of, and theories on organisational structures, organisational behavior, organisational change, and change management;
- describe and conceptualise organisations as embedded in a wider system, and interpret these from a transdisciplinary perspective;
- obtain a deeper understanding of communication from the perspective of sharing and influencing results, and to learn how to deal with conflicts;
- identify and discuss theories on leadership and interpersonal behaviour;
- make an informed choice between statistical techniques commonly used in health and life sciences, applying these to conduct analyses while using statistical software (e.g., SPSS) and interpreting the results of these techniques (e.g., t-tests, chi-square tests) to inform decision making;
- practise analytical and advisory skills.

Course Content

Organisations need to respond and adapt to internal and external challenges. It is increasingly recognized that addressing these challenges is not an ‘isolated process’. Organisations (need to) collaborate with multiple stakeholders in these processes of (continuous) change. In-depth understanding of the structure of an organisation and the wider system it operates in is therefore important. Different forms of knowledge can be used to gain this understanding. It requires insights gained from the analyses of qualitative and quantitative data collected within organisations and their wider context. This course offers insights into these complexities, as well as practical (quantitative) methods to address these.

In interactive lectures, several (guest)speakers will discuss macro-aspects of organisational behavior, including designing organisations and managing them, thereby touching on the managerial skills and the strategic thinking required. In addition, concepts related to motivation, managing interpersonal behaviour, leadership, communication and developing and changing organisations will be explained. Furthermore, lecturers will discuss concepts of systems thinking, and the value of adopting multiple perspectives in analysing organisations and processes of organisational change. The (guest) speakers will illuminate concepts derived from the literature by relating them to their own experiences with informing and managing organisational change in practice. Furthermore, the role of quantitative data in managerial decision making will be explored (the role of qualitative data is covered in other courses). To that end, you will explore and discuss different statistical techniques (e.g., t-tests, chi-square tests) commonly used in health and life sciences. You will conduct analyses with these techniques in statistical software (e.g., SPSS), interpret the results and translate these to decision making processes, working towards an advisory report for a (fictive or real-world) organisation’s management.

In an assignment that runs through the course, practical cases of healthcare companies will be analysed and discussed, resulting in advisory reports for management. With the other students you discuss your experiences and a coach helps you relate the experiences to theory.

Additional Information Teaching Methods

Lectures: approximately 22 hours
Response lectures: 4 hours
Training workshops: 16 hours (attendance is mandatory)
Self-study and writing project assignment: remaining hours.

Method of Assessment

Written exam (60%) and assignment (40%). Grades of both components must be sufficient.

Literature

To be announced on Canvas

Additional Information Target Audience

Compulsory course within the Master programme Management, Policy Analysis and Entrepreneurship for the Health and Life Sciences (MPA) and the Societal differentiation of Health, Life and Natural Sciences Masters programmes

Additional Information

Attendance to training workshops / working groups is indispensable

Custom Course Registration

Please note that Workgroups are formed through Canvas at the start of the course. It is not possible to enrol for workgroups otherwise unless announced via Canvas.

Computational Optical Imaging

Course Code	XMU_0052
Credits	3
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.W.A. Hillmann
Examiner	
Teaching Staff	dr. D.W.A. Hillmann
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130490

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Current Sustainable Energy Technologies

Course Code	X_422582
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.N. Frese

Examiner	dr. R.N. Frese
Teaching Staff	dr. R.N. Frese, dr. A. Baldi
Teaching method(s)	Seminar, Lecture

Course Objective

This course is about asking the right questions and obtaining reliable answers on everything related to energy usage and existing, large scale, renewable energy resources. You will read the book Renewable energy without the hot air, which provides the starting point of the energy transition we are currently bringing about. After reading and commenting on the text, you can explain the basic features of these technologies and make quantitative predictions for implementing each of these technologies. During the course, you will investigate topics of your own choice, focus on their current status, and seek ways to remove hurdles and bottlenecks. After the course, you have an overview on how to investigate the technical, economic and societal feasibilities of all possible forms of sustainable energy, including relevance and positive and negative effects.

Course Content

The course deals with all parts, Part I, II and III, of the book of MacKay, Renewable energy without the hot air, which is free to download at the book website. During the course you read the book of MacKay and annotate and discuss parts with fellow students.

Based on this and your own interests, you focus on the current state of specific topics at general media websites and other sources, from which you formulate several questions. You research topics of interest first, and focus on specific aspects that cannot be found online and are of interest to policymakers or others involved in renewable energy. From this brief investigation, you formulate focused questions with impact and provide a hypothetical answer with only rough calculations. The hypothetical answers should contain well-argued estimates. All questions must be relevant and realistic and will as such be evaluated.

During several workgroups you will learn as a team the methodology and style of reporting, and receive feedback. Subsequently. teams present their findings and you continue to work individually on your own questions including some from a fellow student. Towards the end you will review the written work of another student, and after the feedback on your work is received, you take the comments into account and prepare the final version of your report.

Attendance to all lectures and workgroups is compulsory. All work must be submitted completely and before the deadlines in order to pass this course or to be eligible for a retake. A minimum score applies for parts of the work in order to pass the course. No parts of the course are transferrable to another year.

Additional Information Teaching Methods

This course is both team and self study based, with feedback during workgroups from students and teachers. You are expected to research topics via sources such as general news and media sites, dedicated websites for specific technologies and expert reports on implementation of technology at a specific place. You report multiple times on your research and can discuss impact of your research within workgroups and individually with teachers. An active study approach of an individual student towards asking questions and feedback is essential.

Method of Assessment

The grade is a weighted average of the following items:

1. Formulation of topical researched questions and hypothesis
2. Answers to questions using the correct procedure
3. reading the book and annotating and discussing topics
4. reviewing work of fellow students

A retake is only allowed once, and only for students that have:

1. participated in all workgroup sessions
2. read the entire book with the minimal requested annotations
3. submitted all requested questions and all answers within the deadline with relevant content
4. submitted a review containing relevant content

Note: no parts of this course is transferable to another year

Entry Requirements

This course is for energy related master programs only. Prior knowledge that is expected is general knowledge of energy and power generation, renewable energy and related issues globally. This course necessitate skills like

finding reliable information on the internet, critically assessing claims related to energy and output on websites or other sources. Also students should be able to assess reliable news sites for topical aspects, quick assessing data for relevance and quick reading of reports by municipalities or organizations.

The course is open to all master students within a Chemistry, Physics or Science Business and Innovation program . As well as all students with a bachelor degree in any of these three programs. All other students must have successfully followed courses in mechanics and electricity and magnetism at university level in possession of a physics based working knowledge of energy and energy related topics.

Literature

Literature: David J.C. Mackay, 'Sustainable energy - without the hot air', 2008; available free online at ttp://www.withouthotair.com
Prior the course you can browse the book but do not read it as this is being monitored during the course and part of the grade.

Additional literature that you can read: general media websites on renewable energy, news sites, sustainability and repositories of energy data

Additional Information Target Audience

This is a full time course. Following other courses, including internship in the same period is not recommended. Note: work or parts of work cannot be transferred to another year.

Additional Information

This course is individual research driven and contains reading of the book, and reading of topics of interest on the internet or other sources. Subsequent self-formulating research questions, hypothesis and answers. Each week there are two days of three hour classes.

Explanation Canvas

On canvas most relevant information can be found in the form of examples, presentations and other docs. Students are expected to read carefully all these docs without specific instruction by a teacher. Also teachers will gradually update the canvas site, communicate to all students, specific groups or individual students. All participants are expected to actively check for messages during the course on a daily basis. Also, students can actively engage with teachers via Canvas, including continuous feedback during the course

Recommended background knowledge

This course is challenging for those that do not read much, especially no news sites, and who do not have a general working knowledge on issues related to renewable energy and sustainability in our present time. Also an active attitude towards requesting feedback in workgroups and general classes for discussion is very helpful.

Deep Learning for Medical Image Analysis

Course Code	XM_0151
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. O.J. Gurney-Champion MSc
Examiner	dr. O.J. Gurney-Champion MSc
Teaching Staff	dr. O.J. Gurney-Champion MSc
Teaching method(s)	Study Group, Computer lab, Lecture

Course Objective

- The student can explain the principles and use of neural computing via feed-forward neural networks and, particularly, convolutional neural networks.
- The student can apply deep learning architectures to medical images.
- The student can analyse the performance of different deep learning architectures in the context of a deep learning challenge.
- The student can use their insight in technical aspects of medical imaging to construct networks that solve

specific technical aspects in the imaging pipeline.
- The student can justify the use of different physics aspects in deep learning.
- The student can implement and train a fully connected neural network using NumPy.

Course Content

Artificial Intelligence has proven to be a great tool for helping radiologists and pathologists in diagnosing patients, and, ultimately, selecting the best possible patient-specific treatment. Computers can analyse digital images at an unmet speed and can detect patterns that are missed by medical experts. The development and application of artificial intelligence in medical imaging has sped up due to (1) widely available digital medical images, (2) freely available machine learning tools, and (3) high computing performance and GPU's in particular. This combination has led to applications where computers are highly accurate in detecting patterns, and lesions to support diagnosis and prognosis. AI is used over the entire front of medical imaging, including designing optimal image acquisition schemes, acceleration of imaging, reconstruction of imaging, image enhancement, segmentation and classification.

In this course, we will focus on applying deep learning for digital medical image acquisition, processing and automatic analysis of images. The course will introduce the basic concepts of deep learning. Students will get hands-on experience in using the most common deep neural networks that are used in medical imaging, including convolutional neural networks such as U-net. Ultimately, the core of the course will focus on combining your technical background and knowledge about physics to allow you to apply and enhance deep learning approaches for medical imaging.

Additional Information Teaching Methods

- Lectures
- Computer lab session/practical training
- Presentation/symposium
- Self-study

This course contains some self-study via Canvas, such that the students come well-prepared to the classes.
Deep learning is best learnt by applying it. Therefore, each week contains 1 lecture and 3 computer practicals.
Besides the computer pracitcals, we expect students to work on the exercises at home too.
The classes will be interactive and activating.

We have 3 blocks, each block contains 1 week of predominantly theory and 1-2 weeks of hands-on programming of neural networks.
The final exercise will be presented as a poster to your peers.

Method of Assessment

The course consists of 3 blocks, each with an assignment.

- My first network (35%)
- CNN (25%)
- Image reconstruction (20%)

At the end of the course, a mini-symposium will be held in which the students present the results of Block 3 to each other. The posters and discussion will be evaluated by the TAs and teachers.

- Poster Presentation (20%)

Entry Requirements

- The students are **proficient in Python** programming.
- The students should have an understanding of calculus (derivatives, chain rule, etc.)
- The students should have a basic understanding of medical imaging.
- No prior experience in machine learning is expected/required.

Literature

Literature:

- "Understanding Deep Learning" by Prince --> free download at www.udlbook.github.io

Syllabus:

- On Canvas

Practical training material:

- Will be handed out during the classes

Software:

- Python and Pytorch

Additional Information Target Audience

Recommended:

- Students from the BMTP master
- Students from the Physics master
- any other student that fulfills the prior knowledge

This course is **not** meant for AI MSc students as it overlaps with their core courses on explaining the basics of deep learning. There is a separate "AI for medical imaging" course for them.

Additional Information

Participation in the lectures and practicals is expected; only 1 out of 7 classes and practicals can be missed. Contact the coördinator if you are missing a class.

Participants must prepare the classes via self study.

Explanation Canvas

We are using Canvas

Recommended background knowledge

- The students are proficient in Python programming. They have a working version of Python and a Python editor (e.g. PyCharm) on their laptops.
- The students should have an understanding of calculus (derivatives, chain rule, etc.)
- The students should have a basic understanding of medical imaging.
- No prior experience in machine learning is expected/required.

Electronics and Signal Processing

Course Code	XMU_0069
Credits	6
Period	P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. G.J. Kuik
Examiner	dr. G.J. Kuik
Teaching Staff	dr. G.J. Kuik
Teaching method(s)	

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130513

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Entrepreneurship in Science and Technology

Course Code	XMU_0066
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. E. Masurel
Examiner	prof. dr. E. Masurel
Teaching Staff	prof. dr. D. Iannuzzi, prof. dr. E. Masurel
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130496

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

From Physics to Physiology

Course Code	X_430112
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I.H.M. van Stokkum
Examiner	dr. I.H.M. van Stokkum
Teaching Staff	dr. I.H.M. van Stokkum
Teaching method(s)	Seminar, Study Group, Practical, Written partial exam, Lecture

Course Objective

- To understand fundamental (bio)physical processes and their relevance for physiology.
- To gain knowledge of the application of physical methods in modelling physiological processes
- To gain knowledge of the most important theoretical and practical concepts in modelling and simulation of

biomedical processes at different scales.
- To gain experience with and to apply MatLab and Mathematica to acquire, analyse and evaluate biomedical signals and to model and simulate biomedical processes.

Course Content

The course starts with a general overview of the various types of models used to describe biomedical processes by parametric and non-parametric models using linear and non-linear (differential) equations. Some of the fundamental biophysical principles that have broad relevance in biomedical research will be presented. Practical problems discussed during the lectures will concentrate on modelling the cardiovascular system using mechanical and electrical analogues.

The lectures during the first four weeks will be combined and followed by working groups in which, through relatively simple exercises, experience will be gained of MatLab and Mathematica, two frequently used 4th generation computer languages.
Finally, students will be offered a choice of 1 out of several modelling problems to be solved in small groups, guided by a supervisor. At the end of the course each group will present and discuss their work with all participants and supervisors of the course.

Additional Information Teaching Methods

Lectures, working groups, assignments.

Method of Assessment

Homework assignments (Pass/Fail), report and presentation on modelling problem (40%) and written exam (60%).

Literature

Syllabus.

Additional Information Target Audience

mBMTP, mPhys, mSBI, mComputationalScience

Laser Cooling and Trapping of Atoms and Molecules

Course Code	XMU_0070
Credits	3
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M. Beyer
Examiner	dr. M. Beyer
Teaching Staff	dr. M. Beyer
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130510

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Light-Tissue Interaction

Course Code	X_428572
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.W.A. Hillmann
Examiner	dr. D.W.A. Hillmann
Teaching Staff	prof. dr. A.G.J.M. van Leeuwen, dr. D.W.A. Hillmann, prof. dr. J.F. de Boer
Teaching method(s)	Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130453

Additional Information Teaching Methods

Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Management of Sustainable Innovation

Course Code	E_DBI_MANSI
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	D. Diriker
Examiner	D. Diriker
Teaching Staff	D. Diriker
Teaching method(s)	Study Group, Lecture

Course Objective

Bridging Theory & Practice (knowledge):
Understand and critically assess theories of organizing for sustainable innovation and understand how entrepreneurs , business leaders, governments, and citizens can help to co-create a more sustainable future and how digitalization might be leveraged to support this future.

Bridging Theory and Practice (application):
Apply theoretical insights to conceive new approaches to organizing the implementation of sustainable innovations.

Professional & Social skills:
Ability to work effectively in a team. Have a good contribution, content and process wise: Pro-active teamwork, timely, friendly and constructive.

Course Content

Sustainability is now at the core of our societies and economies: governments, companies and citizens join forces to fight global warming, plastic pollution, and other negative developments. Rapid changes due to digitalization have further unleashed new forms of organizing that enable and accelerate coordination, collaboration, and knowledge efficiencies.

Despite the promise of digitalization and sustainability imperatives for transforming business and society, a strategic focus on convergence of these imperatives as a means for organizing and accelerating the development and implementation of sustainable innovations is relatively new for both practice and scholarship.

In this course, we explore how business leaders and entrepreneurs around the world do things differently, to radically change the way they do business, what role they want to play in our societies, and leverage digital technologies to overcome barriers to sustainable development. You will apply theories on sustainable innovation, digital innovation, innovation processes to examine contemporary cases and learn how to organize for sustainable innovation by mobilizing and integrating information, knowledge and critical resources of diverse stakeholders.

Additional Information Teaching Methods

Lectures: Interactive Lectures (Online & In-person); Guest Lectures Tutorials: Interactive discussion and case work (Online & In-person)

Method of Assessment

Individual written exam.
Group assignment (exact assignment will be announced later).
(Peer) evaluation of contribution to the course and group

Entry Requirements

For the course it is required that students know:

- The concept of Triple P: Planet, people, profit.
- The Brundtland report Our common future 1987
- The 17 UN Global Sustainability goals.

Literature

Academic articles (to be announced in course manual).

All lectures will discuss 2-3 academic articles that will help students to understand sustainable innovations and the organizing processes that are necessary to create and implement these.

Book for background reading:
Senge, Peter M., Bryan Smith, Nina Kruschwitz, Joe Laur, Sara Schley, 2008, The Necessary Revolution: How Individuals and Organizations Are Working Together to Create a Sustainable World, Doubleday (or later edition)

Additional Information Target Audience

Students that want to make a change in existing companies, or co-create change in our societies. This is a course in which students are requested to participate very actively! Students that do not have sufficient time or motivation, are advised not to follow this course.

Concrete profile:

- Students that want to become business or policy consultant in sustainability.
- Students that want to work in a community or province to develop knowledge on how regions can be transformed to co-create value in business and society.
- Students that want become social entreprenuers or implement sustainable innovations in their future careers

Recommended background knowledge

Knowledge of innovation management, diffusion of innovation.
Please note: This course is not about innovation (technical) but about the management of innovation, e.g. stakeholder management, co-creation, open innovation, sustainable entrepreneurship.

Managing Science and Technology in Society

Course Code	AM_470586
Credits	6
Period	P1
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.P. van Oudheusden
Examiner	dr. M.P. van Oudheusden
Teaching Staff	dr. M.P. van Oudheusden
Teaching method(s)	Study Group, Lecture

Course Objective

In this course, students:

- acquire knowledge and understanding of philosophical and social science theories about the mutual shaping of science and technology in society
- acquire knowledge and understanding of basic concepts, issues, and approaches in the field of science and technology studies
- acquire knowledge and understanding of ways in which technology can be democratized and made responsive to societal concerns and expectations
- acquire knowledge and understanding of interactive methods to direct and guide developments in science and technology
- learn to connect basic STS concepts to their own life worlds - and vice versa
- learn to communicate verbally and in scientific writing about their knowledge and understanding and to critically reflect on their own perspectives

Course Content

The ‘Managing Science and Technology in Society’ course offers an advanced introduction to the academic field of ‘Science Technology & Society Studies’ (Science and Technology Studies), as part of the second year of the master ‘Management,

Policy Analysis and Entrepreneurship for the Health and Life Sciences’.

As a MPA student you are trained to operate at the interface of your natural science discipline and society, thereby making a contribution to answering the complex problems arising in these areas. At the dawn of the 21st century, technology and science have an enormous potential for transforming life on earth. At the same time, the dimensions of our human culture shape the directions in which science and technology develop. The production of scientific knowledge and technological objects can solve some of our problems, while giving rise to new challenges. During this course you will study the interactions of science and technology with society, and the various ways in which they mutually shape one another. These interactions invoke a lot of questions. Should we embrace big data? How do new human reproductive technologies interfere with the way we deal with sexuality and social responsibilities? How can we innovate responsibly?

In this course you will be introduced to a conceptual framework to critically and reflexively assess these kinds of questions. The course aims at understanding the intertwinement of science, technology and society, and underscores the importance of a broad concern with these interactions, in order to shape collective futures that take into account societal concerns and expectations.

Additional Information Teaching Methods

‘Managing Science and Technology in Society’ is a fulltime course of eight weeks (6 ECTS). The course schedule is available on Canvas. The total study time is 168 hours. Tuition methods include lectures, work groups, a group project and self-study. The different elements have the following study time: lectures 22 hours; work groups 12 hours; group project 32 hours; self study (including mini-essays) 86 hours; examination (on-campus) 2.5 hours.

Method of Assessment

The examination consists of:

- Mini-essay 1 (10%)
- Mini-essay 2 (10%)

- Final essay (take-home essay exam) (50%)
- SCOB-project (30%)

Both the essay exam and the SCOB project need to be passed.

Literature

The reading material for this course can be found in Canvas.

Additional Information Target Audience

This is a compulsory course for second year students of the master Management, Policy Analysis and Entrepreneurship in the Health and Life Sciences. The course is optional for other Master students in the health and life sciences.

Materials for Light-Energy Conversion

Course Code	XMU_0071
Credits	3
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Ramanan
Examiner	dr. C. Ramanan
Teaching Staff	dr. C. Ramanan
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130498

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Parameter Estimation Applied to Medical and Biological Sciences

Course Code	X_432631
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science

Course Coordinator	dr. I.H.M. van Stokkum
Examiner	dr. I.H.M. van Stokkum
Teaching Staff	dr. I.H.M. van Stokkum
Teaching method(s)	Practical, Lecture

Course Objective

In this course the student learns how to estimate the parameters of a mathematical model that describes the measurements. The statistical theory is illustrated by simulations of parametric models, and practised by solving inverse problems. The goal of the course is to provide insight into the theory of parameter estimation and to develop a critical attitude towards its application and interpretation in order to avoid inconsistent and improper use of the theory.

Course Content

Case studies are taken from medical and biological sciences, in particular describing the dynamics of molecular systems with the help of time-resolved spectroscopy and brain imaging kinetics. Linear and non-linear regression analysis are treated, as well as confidence intervals and significance testing. Topics are Occam's razor, the maximum likelihood principle, linear and non-linear models with single or multiple responses, experimental design, orthogonal polynomials, separable nonlinear least squares, and the Singular Value Decomposition.

Additional Information Teaching Methods

Lecture and exercises with RStudio.

Method of Assessment

Report of the practical exam (100%).

Entry Requirements

Statistics, linear algebra and calculus on BSc level.

Literature

A syllabus will be provided by the lecturer.

Additional Information Target Audience

mBMTP, mPhysics and Astronomy, mComputationalScience

Photosynthesis and Energy

Course Code	X_422553
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. V.U. Chukhutsina
Examiner	dr. I.H.M. van Stokkum
Teaching Staff	dr. V.U. Chukhutsina, dr. I.H.M. van Stokkum
Teaching method(s)	Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130439

Additional Information Teaching Methods

Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Research Methods for Analyzing Complex Problems

Course Code	AM_1182
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	drs. D.H.J. Lynch
Examiner	drs. D.H.J. Lynch
Teaching Staff	A. el Kahal, M.S. van der Loo MSc, B.R. Chakmak, E. Blokbergen, J.R.S. Tjien Fooh MSc, A.R. de Leeuw, K.E. Mujica Gastelbondo MSc, M. Fagan, L.L. Ruijsch, H.B.A. Tahboub MSc, S.S. Mokashi, D.P. Warnaars, A.E. Wilms, D.M.E. Servania MSc, J. Putter MSc
Teaching method(s)	Computer lab, Seminar, Written partial exam, Lecture

Course Objective

The objectives of this course are:

- To understand the differences between beta- and gamma research;
- To acquire insight and understanding of a real-world research process, including knowledge of the character of complex societal issues and the needs, advantages and disadvantages of real-world research;
- Being able to create an adequate research design for the investigation of a specific complex societal problem.
- To acquire insight into relevant research methods (both quantitative and qualitative) to address complex societal problems, their underlying theoretical concepts and their relative strengths and weaknesses;
- Being able to apply these various research methods in a specific societal context;
- To analyze and interpret qualitative findings;
- To acquire insight into reflexivity on the process of doing research in the real-world

Course Content

Contemporary societies increasingly face complex social problems, such as climate change, HIV/ AIDS or ethnic and religious diversity. These complex problems involve a variety of social actors: policy-makers, professionals, NGOs, industries, science and, of course, the public at large. Addressing these complex issues demands an approach that investigates, analyzes and integrates the positions and knowledge of different actors.

This course offers an (advanced) introduction to various research methods used in real world research, including questionnaires, surveys, semi-structured interviews, and focus groups. These methods are commonly used in research into complex problem contexts, communication and opportunities for intervention. Strengths and weaknesses of each research method and technique will be discussed, as well as its possibility to be applied in different societal contexts.

Additional Information Teaching Methods

Research Methods for Analyzing Complex Problems is a parttime course of eight weeks (6 ECTS). The total study time is 160 hours. Tuition methods include lectures, workgroups, workshops, group project work and self-study. The different elements have the following study time: - lectures 20 hours - workgroups and training 36 hours - examination 2 x 2 hours - project work & reading (self-study) Remaining hours Please note that attendance to the workgroup sessions is compulsory. If you miss one workgroup, with a good reason, you will receive an additional assignment. If you miss more than one workgroup session it is no longer possible to pass the project part of the course. Attendance to the lectures is highly recommended. In our experience, relying on self-study alone is insufficient to apply the theory of the lectures in the assignments of the workgroups, and to pass the exam.

Method of Assessment

The course grade is based on the group assignment 'research design' and two individual exams.
Individual Exams 60% of final grade
Group assignment 'research design' 40% of final grade

Literature

The literature of this course consists of selected scientific articles that are provided on Canvas, and the books:

- Verschuren, D.E. and Doorewaard, H. (2010). Designing a Research Project (2nd edition)Eleven International Publishing, the Hague. ISBN 978-90-5931-572-3.
- Gray, D.E. Doing Research in the Real World (3rd or 4th edition) Publications Ltd, United Kingdom. ISBN 978-1-4462-6019-7

An overview of additional literature per lecture will be provided on Canvas.

Additional Information Target Audience

The course ‘Research Methods for Analyzing Complex Problems’ is a compulsory course for first year master students ‘Management, Policy Analysis and Entrepreneurship in Health and Life Sciences’. This course is also a compulsory course within the Majors 'Science Communication' and 'Science in Society' of Health, Life and Natural Sciences Master programmes.

It can be followed as an optional course for other Life Sciences Master program students at the VU University, but only when taken in parallel with Analysis of Governmental Policy (AM_470571). Please contact the course coordinator prior to enrollment if you belong to this group.

Additional Information

The entire course is in English.

Please note that attendance to all workgroup sessions is compulsory.

Attendance to the lectures is highly recommended. From our experience, relying on self-study alone is insufficient to apply the theory of the lectures in the assignments of the workgroups, and to pass the exam.

Contact: Durwin Lynch (d.lynch@vu.nl)

Custom Course Registration

Please note that Workgroups are formed through Canvas at the start of the course, based on the actual number of students that start with the course. It is not possible to enrol for workgroups otherwise - unless announced via Canvas.

Science and Communication

Course Code	AM_470587
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Boersma MA MSc.
Examiner	dr. C. Boersma MA MSc.
Teaching Staff	dr. C. Boersma MA MSc., T.F.L. Roedema MSc, L.B. Clauss MSc, K.E. De Long MSc
Teaching method(s)	Study Group, Lecture

Course Objective

- Gain theoretical insight in the nature of science,
- Gain theoretical insight in the nature of communication,
- Gain theoretical insight in the relationship between science and society,
- Gain insight in the role of science communication in this relationship,
- Acquire knowledge of different theories and models of science communication,
- Acquire knowledge of different strategies, media and activities for science communication,
- Learn how to practically apply theoretical concepts from the field of science communication in communicating

science,
- Develop practical skills for science communication (especially writing and giving oral presentations).
- Reflect on your own knowledge and competencies pertinent to your projected (ideal) role as science communicator.

Course Content

Science is all around us and shapes our lives in many different ways. From the vaccines you need to get when traveling abroad to the smartphone you use on a daily basis, and from the public transportation you use to get to the university to the ingredients of your toothpaste: scientific knowledge is elemental to all of these. Simultaneously, society shapes the ways in which science and technology develop too. Science, technology and society influence each other continuously—or, to put it differently, they ‘communicate’.

Students of the Science Communication specialization are expected to become experts in understanding and designing interactions between science and society. In order to make this interaction fruitful and valuable for both science and society, it is first of all important to gain theoretical knowledge about science, about communication and about science communication. Science and Communication provides students with the theoretical and conceptual foundations of the discipline of science communication. Thus, you will develop an in-depth understanding of communication processes at the core of several interfaces, including those between scientists from different disciplines, between different sciences and their stakeholders, and between science and the public.

Additional Information Teaching Methods

Expected contact hours:
Lectures (20 h)
Workgroups (25 h)
Home-study for group assignments (15 h)
Home-study for individual assignments/exam (100 h)

Method of Assessment

a) Participation. (pass/fail)
This consists of the following:
- the 'CARQ' presentations and summaries;
- Active participation in the work groups.
All these are assessed as pass or fail. For each one you fail, you have to do an alternative assignment.
Nota bene: if you fail your participation, this cannot be compensated with an alternative assignment!

b) An interview with a science communicator of your own choosing, including a reflection report. (20%)
d) "TED-talk" in which you present the research you did (e.g. for your Bsc thesis or (first) Msc internship) and reflect on it. (20%)
e) Exam. (60%)

To pass, your grades for the assignments and exam all need to be 5.5 or higher.

Resit: In case your grade is not sufficient, you have to take a resit. This can either consist of a second attempt at (b), (c) or (d), or a re-exam.

Literature

Academic articles. Direct links to articles will be provided on Canvas.

Additional Information Target Audience

The course Science and Communication is a compulsory course for students of the Master specialisation Science Communication (Wetenschapscommunicatie) and is a prerequisite for the internship. Science and Communication is an optional course for students from other master programs in the health and life sciences.

Science and Society in Historical Perspective

Course Code	X_400424
Credits	6
Period	P4+5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science

Course Coordinator	dr. G.J. Somsen
Examiner	dr. G.J. Somsen
Teaching Staff	dr. G.J. Somsen, dr. D.J. Beckers
Teaching method(s)	Seminar, Lecture

Course Objective

To understand how science and society have co-evolved and to know how to research this co-evolution.

Course Content

We are used to thinking of science as the prime shaper of our modern world, but how exactly does that work? How have the sciences come to play this role in our society and, reversely, how have politics and culture impacted the scientific enterprise? In this course we will study the changing relationship between science and society over the centuries by looking at a series of themes. We will examine how scientists have dealt with the public, we will study science's relationship to religion, and we will ask how Western science really is. To address these questions, we will use historical literature, videos and podcasts as well as original source texts and pictures. We will also try our own hand at historical research, through a number of skills trainings and a final paper.

Additional Information Teaching Methods

Seminars (2hrs per week), skills trainings (twice), lectures (occasionally), presentations ands feedback.

Method of Assessment

Assessment is based on active participation during the seminar (10%) and a piece of historical research, reported on in a presentation (20%) and a final paper (70%).

Entry Requirements

Bachelor degree

Literature

Available via Canvas

Additional Information Target Audience

- Master students in the sciences interested in historical reflection;
- Master students in the humanities interested in understanding the role of science in society.

Additional Information

More information with the course coordinator: Geert Somsen, Athena Institute, g.j.somsen@vu.nl

Recommended background knowledge

None necessary, except what you have learned in your BA.

Science in Dialogue

Course Code	AM_1002
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E. Gommeh
Examiner	dr. E. Gommeh
Teaching Staff	mr. D.T. van der Lugt MSc, P.E. van der Aar, M.C. Vos MA MSc., J. Putter MSc, A.M. Aris, A.R. de Leeuw
Teaching method(s)	Seminar, Study Group, Written partial exam, Lecture

Course Objective

To gain knowledge of and insight into:

- the basic concepts and issues in understanding science-society interactions and interfaces, both from a science and technology studies and communication science perspective
- the nature and course of interpersonal and group communication processes relevant to the formal and informal dialogue between science and society
- the nature and form of dialogical science communication, aimed at reflective learning and mutual understanding

To acquire or improve:

- individual skills for effective interpersonal communication
- individual skills for the design and facilitation of science-society dialogue

Course Content

This course invites you to explore the democracy of science and technology with a particular focus on the communicative aspects of a fruitful science - society dialogue. Science, technology and innovation hold a great promise for the progress of our societies. Take for example the opportunities of nanomedicine for targeted drug delivery or synthetic biology for renewably energy or the recycling of plastic waste. At the same time these developments are potentially controversial. They lead to a variety of questions and concerns related to risks, benefits and wider moral issues. For example, nanotechnology may contain risks for health and environment. Synthetic biology may radically change the nature and meaning of life.

Clearly, advances in science do not always match the needs, desires and expectations of stakeholders and citizens in society. On the other hand, parts of society might not always appreciate the nature of scientific findings and what they have to offer. For a fruitful relationship between science and society – especially regarding topics that are potentially controversial – a constructive science-society dialogue is necessary.

This course offers advanced lectures on the basic concepts and theories about the dynamic relationship of science and society, the nature of societal controversies surrounding science, technology and innovation, and the basic concepts and issues of dialogical science communication: communication, learning, dialogue, understanding, controversy, and democracy. A series of seminars and group assignments present communicative tools and interaction spaces that can be used to design and facilitate science-society interactions. Through training workshops you will focus on improving your individual communication and facilitation skills.

Additional Information Teaching Methods

Lectures (14h), Workgroups (28h), Training workshops (24h), Dialogue presentations (12h), Selfstudy (remaining hours)

Method of Assessment

Group assignment (50%), Two digital exams (30%), Reflection report (20%).

All assignments must be passed (grade > 6).

Literature

Is announced on Canvas one month before start of the course

Additional Information Target Audience

Optional course in the MSc specialization Science Communication

Additional Information

Independence and a cooperative attitude is expected. Attendance to workgroup/ seminars and training sessions is mandatory.

Science Journalism

Course Code	AM_471014
Credits	6

Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Boersma MA MSc.
Examiner	dr. C. Boersma MA MSc.
Teaching Staff	A.M. Aris, K.E. De Long MSc, dr. C. Boersma MA MSc., T.F.L. Roedema MSc
Teaching method(s)	Computer lab, Study Group, Lecture

Course Objective

After following this course, you are able to:

- Review the key changes in science journalism due to changes in the science-society relationship and digitalization of the media
- Examine contemporary issues of science journalism including the emergence of issues and publics, framing, values and emotions, expertise and evidence, uncertainty and ignorance.
- Apply an interdisciplinary toolkit to think about the interactions between science, media and the public and the role of science journalism
- Useyour skills in different science communication media, such as writing, video, and documentary to generate meaningful exchange between science and society
- Describe what the professional practice of science journalism entails

Course Content

Science journalism is needed now more than ever. Science heavily influences our society, not only through the application of its knowledge but also through its worldview. Although science and technology have shown the potential to improve our lives and living conditions significantly, they also contribute to new problems, risks and concerns. This awareness has changed the science and society relationship and put science and technology under increasing public scrutiny. In the context of the complex challenges of today, such as climate change or food security, ‘the facts’ are regularly called into question. Public trust in the solutions presented by science is under pressure. There is clear demand for the scientific world to communicate effectively with the rest of society. But how to shape the public conversation about science in society? And how do the media shape our perception and appreciation of science? What is the role of science journalism? Especially with respect to the larger transformations that are taking place in the relationship between science, media and public. Digitalization has revolutionized the media landscape. Everyone now has the means to either find or generate any information themselves. This provides new opportunities, but may also spread low-quality, sensationalized and sometimes intentionally misleading information. The challenge for science journalism today is to reinvent its practice and help shape a meaningful exchange between science and society.

In this course we review the key changes in science journalism. We examine how changes in both the science-society relationship and digitalization of the media landscape have changed the science journalism practice. We look at different roles of journalism such as reporters, storytellers, watchdogs and educators and discuss what new roles might need to be developed in light of the abovementioned changes. We analyze the communication and interaction dynamics in the current media landscape according to a set of core questions: How do issues and publics emerge? What is the role of framing? What is the role of values and emotions? What is the role of expertise, evidence? What is the role of uncertainty and ignorance? Finally, we train specific skills that you need as a science journalist, such as popular writing, making popular science videos, interviewing, and program design. Throughout the course, we work together closely with professionals from the field. Our guest speakers work as freelancer, editor or producer at diverse science media, such as newspapers, broadcasting media, online magazines and documentary film. They contribute to our course via the lectures, assignments and discussions.

Additional Information Teaching Methods

Lectures and seminars on theory and practice of science journalism and skill training (36h). Considerable time is set aside for performing science journalism in assignments (108h). All sessions indicated as training sessions in the schedule are mandatory. The assignments are assessed by lecturers and fellow students (peer-review process). Self-study (remaining hours).

Method of Assessment

Several individual assignments (60%), several small group assignments (40%).
All assignments must be passed (grade > 6).

Literature

Announced on Canvas one month before start of the course

Additional Information Target Audience

All Master students with a Beta-Bachelor degree. Students taking this course as part of their VU Science Communication Specialisation or UvA Major Science Communication will have precedence over other students. Students from other faculties and or universities need to get formal consent from the course coördinator (Frank Kupper) before enrolment.

Additional Information

The course is taught in English. Because language proficiency is crucial, there is the option to hand-in assigments in Dutch. More information: t.f.l.roedema@vu.nl.

Scientific Writing in English

Course Code	X_400592
Credits	3
Period	P3, P6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	B.I.M. Schouten
Examiner	B.I.M. Schouten
Teaching Staff	
Teaching method(s)	Study Group

Course Objective

The aim of this course is to provide Master's students with the essential linguistic know-how for writing a scientific article in English that is well organized, idiomatically and stylistically appropriate and grammatically correct.
At the end of the course students:

- know how to structure a scientific article;
- know what the information elements are in parts of their scientific article;
- know how to produce clear and well-structured texts on complex subjects;
- know how to cite sources effectively;
- know how to write well-structured and coherent paragraphs;
- know how to construct effective sentences;
- know what collocations are and how to use them appropriately;
- know how to adopt the right style (formal style, cohesive style, conciseness, hedging)
- know how to avoid the pitfalls of English grammar;
- know how to use punctuation marks correctly;
- know what their own strengths and weaknesses are in writing;
- know how to give effective peer feedback.

Final texts may contain occasional spelling, grammatical or word choice errors, but these will not distract from the general effectiveness of the text.

Course Content

The course will start with a general introduction to scientific writing in English. Taking a top-down approach, we will then analyse the structure of a scientific article in more detail. As we examine each section of an article, we will peel back the layers and discover how paragraphs are structured, what tools are available to ensure coherence within and among paragraphs, how to write effective and grammatically correct sentences and how to choose words carefully and use them effectively.

Topics addressed during the course include the following:

- Structuring a scientific article
- Considering reading strategies: who is your readership? How do they read your text? What do they expect? How does that affect your writing?
- Writing well-structured and coherent paragraphs
- Composing effective sentences (sophisticated word order, information distribution).
- Academic tone and style: hedging – why, how, where?

- Using the passive effectively
- Understanding grammar (tenses, word order, etc.)
- Understanding punctuation
- Referring to sources: summarising, paraphrasing, quoting (how and when?)
- Avoiding plagiarism
- Vocabulary development: using appropriate vocabulary and collocations

Additional Information Teaching Methods

Scientific Writing in English is an eight-week course and consists of 2 contact hours a week. Students are required to spend at least 6 to 8 hours of homework per week. They will work through a phased series of exercises that conclude with the requirement to write several text parts (Introduction, Methods, Discussion and Abstract). Feedback on the writing assignments is given by the course teacher and by peers.

Method of Assessment

Students will receive the three course credits when they meet the following requirements:

- Students hand in three writing assignments (Introduction, Methods, Discussion)
- Students get a pass mark for all writing assignments;
- Students provide elaborate peer feedback (Introduction, Methods, Discussion, Abstract);
- Students attend at least 7 out of 8 sessions;
- Students are well prepared for each session (i.e. study the theory for each session and participate in online discussions about the theory);
- Students participate actively in class and respect deadlines;
- Students do not plagiarise or self-plagiarise.

Writing assignments:

- If students have a BSc thesis in a traditional thesis form (e.g., 20+ pages) and written in English, they may use this for the writing assignments.
- If students have a BSc thesis in a traditional form (e.g., 20+ pages) written in another language than English, they may use this for the writing assignments.
- If students have written a paper or report in English that’s not already in article form, they may use this for the writing assignment.
- If students are working on their MSc thesis or internship report when taking Scientific Writing in English, they may use this for the writing assignments. They will have to notify their supervisor to make sure that they won’t be accused of self-plagiarism.
- If students cannot or do not wish to use any of the above-mentioned texts for the writing assignments (1-4), they are expected to do a limited Literature Review on a topic in their field of research, using at least 5 articles.

Students are not allowed to use the following texts for the writing assignments:

- A BSc thesis written in English that’s already in article form.
- A MSc thesis written in English that’s already in article form (and that has already been marked).
- An internship report written in English that’s already in article form (and that has already been marked).
- A paper or report written in English that’s already in article form.

Literature

Effective Scientific Writing: An Advanced Learner's guide to Better English, 5th edition (May 2017) (A. Bolt & W. Bruins, ISBN 978 90 8659 617 1). VU bookstore: €34.50

Additional Information Target Audience

This course is only open to students of the two-year Master's programmes of the Faculty of Science. These students are only eligible to the course if they have already conducted scientific research (e.g. for their Bachelor’s thesis) or if they will be working on a research project when taking Scientific Writing in English.

Additional Information

To do well, students are expected to attend all lessons. Group schedules are to be found at rooster.vu.nl and on Canvas.

A VUnet registration for this course automatically gives access to the corresponding Canvas site. Group registration only takes place via Canvas.

Make sure Scientific Writing in English does not overlap with another course.

If you have registered for a group in Canvas, you are expected to attend all sessions (eight). If you decide to withdraw from the course, do so in time in VUnet. This will avoid a 'fail' on your grade list for not taking part in this course and allows other students to fill in a possible very wanted group spot.

Full time students with their main registration at VU will be given preferential treatment for placement in this course. For secondary students proof of enrollment is not a guarantee of placement.

Custom Course Registration

Important: each group has a minimum of 18 and maximum of 24 participants, so students should register on time through VUnet to ensure a place in one of the (designated) groups. If you have registered for a group in VUnet, you are expected to attend all sessions (eight). If you decide to withdraw from the course, please do so in time. This allows other students to fill in a possible very wanted group spot.

Soft Condensed Matter and Biological Physics

Course Code	X_420167
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. ir. G.J.L. Wuite
Examiner	prof. dr. ir. G.J.L. Wuite
Teaching Staff	prof. dr. P.R. ten Wolde, prof. dr. ir. G.J.L. Wuite
Teaching method(s)	Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130433

Additional Information Teaching Methods

Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Tracer Kinetic Modelling

Course Code	XM_0014
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. M.M. Yaqub
Examiner	dr. ir. M.M. Yaqub
Teaching Staff	dr. S.V.S. Golla, dr. ir. M.M. Yaqub
Teaching method(s)	Seminar, Practical, Lecture

Course Objective

The 6 EC master course Advanced Medical Technology aims to provide the student knowledge and skills in the field of Tracer Kinetic Modelling. Starting with the basis of PET-imaging and signal-&-system-analysis the courses develops to an state-of-the-art discussion of

1. different tracer-kinetic models and their physiological interpretation,
2. signal-&-system-analysis based administration (e.g. bolus infusion & steady state) of radiotracers and radiotracer concentration measurements,
3. estimation of model parameters using different least square estimation techniques.

Course Content

This year the course Advanced Medical Technology concerns the topic of Tracer Kinetic Modelling applied to Analysis of PET studies. To be short, a PET image visualises the spatial distribution of a radio-pharmaceutical (or radiotracer i.e., radio-labelled substances) within the human body. The quantitative analysis of the radiotracer distribution and its temporal behaviour, using the TracerKinetic-Modelling approach, enables the interpretation of the PET studies in terms of physiological processes, such as regional perfusion of the myocardium and brain, glucose utilisation in tissues and tumors, and binding of medicines at receptors in the cell membrane. Topics highly value for clinical diagnostics and, moreover, topics in state-of-the-art fundamental and clinical research. Tracer Kinetic Modelling is the field of studying the time course of a radiotracer and its metabolites in the human body by using quantitative methods such as: the Positron Emission Tomograph (PET), mathematical modelling of pharmacokinetics (using differential equations and the theory of signals & system), estimation of model parameter (using linear and non-linear least-square techniques) and, finally, the tools developed for the tracer-kinetic-model-based analysis and the interpretation and the clinical use of PET-images.

Additional Information Teaching Methods

The course consists of lectures and hands-on computer practicals with assignments and a final exam.

Method of Assessment

Hands-on practicals with assignments & written examination. No scoring is given for the hands-on practicals with assignments but students need to attend all hands-on sessions in order to be allowed to attend the exam. The final score for the course is based on just the exam score.

Explanation Canvas

All important information is communicated through canvas announcements.

Recommended background knowledge

Good to have read about medical imaging techniques before hand (bachelor level). However all needed theory will be explained during the course.

Tutoring Students

Course Code	X_432625
Credits	3
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	I.T. Braadbaart MA
Examiner	prof. dr. J.E. van Muijlwijk-Koezen
Teaching Staff	dr. M. Wijtmans, prof. dr. J.E. van Muijlwijk-Koezen, I.T. Braadbaart MA, dr. D.J. Scholten
Teaching method(s)	Lecture

Course Objective

This course aims to prepare students for coaching tasks in tutorials and practical courses. Students will explore different aspects of teacher-student interaction, and by the end of the course will be able to reflect critically on their beliefs and practices about teaching, guide pre- and post-experiment discussions, support students in problem-solving, plan for safety in practical work, and give written feedback.

Course Content

The course asks students to engage in literature, discussion and practical simulations to support students' understanding of appropriate teaching and coaching practices in tutorials and practical courses. Students will analyse various learning aims within a laboratory setting and prepare, conduct and reflect on a presentation of a pre- and post-experiment discussion in tutorials and practical courses. They will observe and interpret the application of problem solving and coaching models in tutorials and practical courses. Attention will be paid to strengths and weaknesses in models of teacher-student interaction, with a focus on questioning. An important constituent is the student's analysis of and reflection on his/her own teaching approach. Topics on safety and lab journal procedures in practical courses as well as on the grading of lab reports are also included.

Additional Information Teaching Methods

3,5 consecutive hours per session (2x per week over four weeks):

- Lectures and whole class discussions
- Hands-on practice moments
- Self-study
- Group work

Method of Assessment

A collection of written group assignments completed throughout the course, including:

- a summary of good guiding practices
- two lesson plan designs
- a plan for pre- and post-experiment discussion
- a rubric

A collection of individual assignments completed throughout the course, including:

- a reflection on previous learning experiences
- a reflection on pre- and post-experiment discussion
- a reflection on attitudes towards guided- and open-inquiry labs
- an analysis of various strategies to repair student misconceptions in the sciences
- a reflection and analysis of problem-solving approaches
- a final critical reflection on your development throughout the course, accompanied by a formative presentation and feedback session

Literature

Will be provided.

Additional Information Target Audience

mDDS, mChem, mPhys, mMath, mMNS, mSBI

Additional Information

This course is recommendable for any MSc student who has a general interest in educational coaching strategies and models. Specific attention is given to guiding experimental / laboratory work and problem-solving skills.

Number of participants is limited to 24 (first-come, first-serve basis).

Custom Course Registration

VUnet

Explanation Canvas

Canvas will be used to share literature, submit assignments, and for other relevant course communications.

Ultrafast Laser Physics

Course Code	XMU_0035

Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. K.S.E. Eikema
Examiner	
Teaching Staff	dr. P.M. Kraus, prof. dr. K.S.E. Eikema
Teaching method(s)	Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130444

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.